PEER REVIEW HISTORY

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ARTICLE DETAILS

TITLE (PROVISIONAL)	A systematic review on the instruments used for measuring the association of the level of multimorbidity and clinically important outcomes
AUTHORS	Lee, Eng Sing; Koh, Hui Li; Ho, Elaine Qiao-Ying; Teo, Sok Huang; Wong, Fang Yan; Ryan, BL; Fortin, Martin; stewart, moira

VERSION 1 – REVIEW

REVIEWER	Ge Lixia National Healthcare Group, Singapore
REVIEW RETURNED	18-Sep-2020

GENERAL COMMENTS	This systematic review provided a list of instruments for measuring level of multimorbidity and described the requirements, strengths and limitations of these instruments. The SR may benefit from higher level of data synthesis. Here are some major comments: Methods: 1) Articles published from Jan 2010 to Aug 2018 were included. It will be good to conduct an update searching for articles published rom Aug 2018 to Sep 2020 and includes new articles identified in this SR. 2) What are the reasons why only included studies with patients from primary care or general population only? Are there any differences in definition or measure of multimorbidity between hospitalised /institutionalised patients and primary care patients/general population? 3) The Modified Newcastle-Ottawa Scale was used for quality assessment for Cohort Studies, how about for Cross-sectional studies? 4) The purpose of quality assessment of the primary studies is to minimize bias and ensure the findings are reproducible before including these studies in systematic review for high-level evidence generation. However, this systematic review focuses on the measure of multimorbidity instead of the findings/ conclusions from the respective study. Not only the quality of the primary studies is important, but also the psychometric properties of the measure. A study assessed to be high quality does not mean the developed/used measure of multimorbidity is high quality or scientifically sound, whereas a study assessed to be low quality does not mean the developed/used measure of multimorbidity is low quality. To assess the quality of an instrument/measure, different criteria should be used, such as psychometric properties of the measure, meaning the reliability and validity of measure should be evaluated and assessed. The inclusion of outcome measures can serve as a form of validity, but the purpose of reporting outcome measures was not mentioned by the authors. 5) Page 7 Line 38-40, how was hand-searched for the additional

	eligible papers identified in the reference lists of included articles conducted? Please state the database/website used for the searching. Discussion: 1) The first paragraph of the Clinical Implication mentioned the study design of the reviewed studies. How will the type of study design influence the measure of multimorbidity? 2) Why "the definition of high prevalence and significant impact would depend on the type of healthcare setting or geography?" If it is so, will the authors consider recommending different multimorbidity measure for different healthcare setting? Minor comments: 1) Page 7 Line 17-19: We included instruments that measured multimorbidity for the purposes of establishing prevalence as well as instruments that measured multimorbidity for the purposes of predicting outcomes. Suggest changing "as well as instruments that measured multimorbidity for the purposes of" to "and". 2) Page 7 Line 51, suggest changing "reported only instruments" to "only reported instruments". 3) Page 8 Line 24-26, suggest deleting (i.e., cross-sectional and cohort), since the two methods were mentioned earlier.

REVIEWER	Andrew Robert Gray Biostatistics Centre, Division of Health Sciences, University of Otago, New Zealand
REVIEW RETURNED	07-Oct-2020

GENERAL COMMENTS	This interesting systematic review examines contemporary instruments for measuring multimorbidity (the literature search was restricted to January 2010–August 2018), providing both a catalogue of these (from the abstract: "The main objective of this systematic review was to provide a list of instruments for measuring level of multimorbidity") along with assessments of each ("The secondary objective was to provide details of the requirements, strengths and limitations of these instruments"). Such a review would be useful to researchers, refreshing some of the information in Huntley, et al. (2012), particularly for readers who are not specialists in this area. In order to help this group of researchers, I think that the systematic review needs to be refreshed. The searches were conducted over two years ago now and as you state in the registration "There will be the novel ones that have only been recently developed.", a statement that is presumably as true now as it was in 2018. Similarly, in the background (Page 6, Line 15), you suggest exponential growth in the literature, suggesting that there could be a reasonable quantity of work published in those two years. It's not clear to me how the risk of bias is relevant here, and consequently why only studies with a low risk of bias were included in the review. It seems that 67 studies were identified and a third of these (21 studies) excluded due to a higher risk of bias; which in turn excluded one sixth, 6 out of a possible 33, of the instruments—non-trivial reductions in both studies and instruments. As Huntley, et al. (2012) noted, "we have not set out to assess individual study quality", which would have been even more relevant to them as they were interested in both reliability and validity. If you were looking at the results of the studies as your focus (which is not part of the objectives, as I read them), risk of bias assessment would be

	expected, but an instrument is not made less interesting because the researchers used inappropriate statistical tests or did not conceal allocation appropriately (which would increase the study's risk of bias and increase the chances that it was excluded on this basis here, irrespective of the quality of the instrument itself). Can you explain why the risk of bias assessment is needed here? You also refer to the study needing at least one outcome (e.g. Page 3, Line 30; Page 6, Line 56; this is most clearly stated on Page 6, Lines 25–31: "We specifically looked at instruments that measure the prediction of a specific outcome with different levels of multimorbidity in community-dwelling individuals."), and this seems to refer to a dependent variable modelled using the multimorbidity instrument as an independent variable. While the ways that multimorbidity instruments are used and when they were statistically significantly associated with outcomes of interest (as shown on Page 8, Lines 37–43 and in Tables 1, 3a, and 3b) is definitely interesting, although the strength/clinical relevance of the associations would need to be described if this was intended to assess instrument's validity, statistical significance alone would not suffice, if the present work is to serve as a catalogue guiding researchers in selecting an appropriate instrument to measure multimorbidity (as I interpret the study objectives), this seems an unnecessary exclusion. The approach taken here would also seem to rule our prevalence studies. Do you think that different measures of multimorbidity should be used depending on whether the outcome of interest is, for example, quality of life-based versus healthcare cost-based? That would suggest that there are multiple multimorbidity constructs that vary by outcome (not just by population of interest). I think that there is much more to what you say on Page 21, Lines 36–42 than readers might appreciate. Do you believe that the multimorbidity is a single construct, at least for a particular well-defined population, or are there multiple constructs even within a particular well-defined population? Similarly, why would it matter whether multimorbidity was "the main independent variable" and not "a covariate" (Page 7, Lines 5–9)? How would this be operationalised when a study investigates a range of potential variables where one of those was multimorbidity, alongside SES and ethnicity, say, and where all independent variables are of interest? Why would multimorbidity not be of interest if it was a mediator instead of the predictor of interest (say, looking at ethnicity differences in support for euthanasia and asking whether these are mediated by the extent or presence of multimorbidities)? This would seem to potentially penalise some more sophisticated or comprehensive studies, at least in theory, whereas simpler or piecewise publications of the same research could still be eligible. What about multimorbidity as an outcome? What I'm asking with all of these questions is perhaps more simply asked as whether you believe that multimorbidity should be measured in different ways for these different contexts? If so, it would seem more useful to me to stratify the reporting by these criteria rather than excluding studies and focusing on a rather specific use case. It seems very likely to me that readers interested in your work will wish to incorporate multimorbidity in a wide range of contexts (exposure, confounder, mediator, effect modifier, or outcome) and that by restricting your inclusion in this way, you are

	substantially reducing the possible use of your work. If, however, you are only interested in your work being of direct interest to the particular use case defined through your inclusion and exclusion criteria, that should be made very explicit (including perhaps in the title, certainly in the abstract) so that readers with other interests can continue their search elsewhere. If I was a researcher considering instruments for assessing multimorbidity, for any purpose, reliability and validity (as Huntley, et al. looked at) would seem crucial. While you provide information relevant for some forms of validity, I didn't see any direct mention of the issue of reliability. Can you explain why this aspect of the instruments was not included in your review? Reliability coefficients would potentially be amenable to meta-analysis, which could be very helpful to readers. In any case, I think that these should fall under the heading of strengths and weaknesses of the instruments. The burden on participants, where appropriate (interviews) and reported, could be a useful addition to Table 2 for someone weighing up various instruments. The financial cost (see Page 20, Lines 13–17) of instruments would also seem to be of interest and could be easily added to Table 2. In the registration you say that "We expect high heterogeneity in the selected studies and meta-analysis will not be attempted." but it's not clear to me how you would perform a meta-analysis here unless it was looking at reliability or at associations between multimorbidity and outcomes. However, the latter is not part of the study objectives based on my reading. What would such a meta-analysis have looked like otherwise, assuming that heterogeneity was not an issue? I'm not clear as to why non-English language articles were excluded. Could you explain, including in the manuscript, why this approach was taken? There are a few minor issues with language (e.g. "Studies published between January 2010 and August 2018 inclusive, in English language." in the abstract is not a complete sentence. Careful proofreading throughout should find and correct these and other small issues. In the article summary, you say "This review builds upon Huntley et al.'s 2009 review paper". Do you mean their 2012 paper? Their search was up to December 2009, which might be your intended point here? See also Page 6, Lines 12–16 and Line 19, and perhaps elsewhere. You mention (Page 5, Line 45) "a Delphi panel of experts" but Delphi is a method/process not a characteristic. You say that you contacted authors where additional information or clarification was needed (Page 7, Line 57). How often did this occur? How often was it successful? Page 8, Lines 21–27: I would think that a study that combined cross-sectional and cohort aspects would be likely to be a cohort study where analyses were also performed on the baseline measurements. If a study (as a study design) follows participants over time, it is a cohort study not a cross-section/cohort study; there may be cross-sectional and longitudinal analyses, but the study itself

	is a cohort study. I disagree that disease count "assumes a linear relationship on an ordinal scale" (Page 19, Lines 13–16). An ordinal scale is defined in terms of ordering not linearity, and any assumption of a linear relationship would not be a function of the instrument itself, but rather a researcher's use of the instrument through treating it as a continuous exposure variable. Depending on the taxonomy for measurement scales being used, counts are absolute in any case (not ordinal, interval, or ratio). There is nothing stopping a researcher from modelling non-linearities (e.g. through splines, higher-order polynomial terms, or fractional polynomials) or (aside from the reduction in statistical power) creating categories based on cut-points. None of these would "…[ignore] the fact that an additional unit of increase in 'disease count' will likely have a diminishing impact." (Lines 15–19). It is important to keep distinct the aspects of the instrument and aspects of how it could be used. I don't think that it is surprising that only observational studies were found (although I'm not sure whether Page 21, Lines 13–18 is referring to pre- or post-exclusions). As multimorbidity was required to be the primary outcome, intervention studies would only be eligible if they manipulated multimorbidity, which would obviously not pass any ethical review. It seems to me that your inclusion and exclusion criteria themselves would prevent any intervention studies from being included. While Line 17–22 might well be true in general, your findings can't reflect this unless they are based on the lack of such studies identified pre-exclusions. I didn't understand the logic of splitting Table 3 into 2 tables (one for positive associations and one for negative or no evidence of associations [not "no associations", see below]). As an example, wouldn't "deficits of perceived general health" (in Table 3a) be loosely the inverse of "general health", perceived health status", or "self-rated health" (in Table 3b column one)? Whether a relationship is positive or negative could also depend on how the dependent variable is coded (satisfaction 1=low and 5=high, versus the same with 1=high and 5=low, versus dissatisfaction with either NRS). Between these two points, I don't personally find this distinction between the two tables to be helpful. Other groupings, perhaps thematic, might be more useful though. I'd suggest caution in the "no association" part of Table 3b, as you are not claiming that there is no association, merely indicating that there was no evidence for such associations (and I'll point out that this is likely in some cases to depend on the other variables included in the model as well as the sample size). In summary: 1) There doesn't seem to be any justification provided for requiring multimorbidity to be the primary exposure of interest. You might have a reason for this position, but it isn't clear to me. If this is part of the design of the study, it needs to be very clear to readers. 2) Similarly, I don't see the reason for the focus (and requirement) for outcome variables unless there is an argument and subsequent discussion about "horses for courses" in multimorbidity measurement.

	3) The risk of bias assessment didn't seem relevant to me. 4) Some discussion around reliability, and more around validity and clinical significance of associations, could be extremely helpful to readers. 5) I think that the search needs to be refreshed to the present date.

VERSION 1 – AUTHOR RESPONSE

Reviewer 1

Methods:

1) Articles published from Jan 2010 to Aug 2018 were included. It will be good to conduct an update searching for articles published rom Aug 2018 to Sep 2020 and includes new articles identified in this SR.

Thank you for the comments and suggestion. We have updated the search to 23rd Oct 2020**.**

2) What are the reasons why only included studies with patients from primary care or general population only? Are there any differences in definition or measure of multimorbidity between hospitalised /institutionalised patients and primary care patients/general population?

Unlike 'comorbidity' where the health outcomes are evaluated in the context of an index condition where specialists play a greater role in management of specific conditions, the majority of patients with 'multimorbidity' are managed by primary care physicians where the identification of an index disease is often not obvious or useful. Hence, our review focused on the general and primary care populations. This has been updated in the Methods section (second paragraph, page 6).

3) The Modified Newcastle-Ottawa Scale was used for quality assessment for Cohort Studies, how about for Cross-sectional studies?

The modified Newcastle-Ottawa Scale was used for both cohort studies (Appendix 2) and cross-sectional (Appendix 3) studies.

4) The purpose of quality assessment of the primary studies is to minimize bias and ensure the findings are reproducible before including these studies in systematic review for high-level evidence generation. However, this systematic review focuses on the measure of multimorbidity instead of the findings/ conclusions from the respective study. Not only the quality of the primary studies is important, but also the psychometric properties of the measure. A study assessed to be high quality does not mean the developed/used measure of multimorbidity is high quality or scientifically sound, whereas a study assessed to be low

quality does not mean the developed/used measure of multimorbidity is low quality. To assess the quality of an instrument/measure, different criteria should be used, such as psychometric properties of the measure, meaning the reliability and validity of measure should be evaluated and assessed. The inclusion of outcome measures can serve as a form of validity, but the purpose of reporting outcome measures was not mentioned by the authors.

We thank the reviewer for raising this important point that was not addressed properly in the manuscript.

The aim of this review was to provide an overall recommendation of the list of instruments that have been appropriately used to determine an outcome. While the emphasis of this review was on the measure of multimorbidity, the risk of bias assessment gave equal weightage for the risk of selection, comparability and outcomes in the study design. We have amended the Background section (5th paragraph, page 6) to state the aim of the study more clearly - *"The main objective of this review was to list instruments for measuring the levels of multimorbidity where risk of bias of the study design was assessed to be low."*

Assessing the quality of an instrument/measure such as the psychometric properties was beyond the scope of this systematic review. We have put this under 'Strengths and Limitations' segment under the Discussion section (2nd paragraph, page 27) - "*This systematic review did not review the validity and reliability of the list of instruments as it was beyond the scope of the intended work. We have, however, included the references of the original papers or validation studies in Table 2 where each of the instrument was described.*"

5) Page 7 Line 38-40, how was hand-searched for the additional eligible papers identified in the reference lists of included articles conducted? Please state the database/website used for the searching.

We used hand-searching or a manual process of scanning the reference lists of all the selected articles for additional eligible papers, in case they were missed during indexing and ensure relevant articles are not overlooked. We obtained the abstract and full text by using PubMed Central website from the U.S. National Institutes of Health's National Library of Medicine.

Discussion:

6) The first paragraph of the Clinical Implication mentioned the study design of the reviewed studies. How will the type of study design influence the measure of multimorbidity?

The type of study design will not influence the measure of multimorbidity. We have clarified the sentence in the Discussion section under 'Clinical Implications' segment (1st paragraph, page 26). *"The lack of interventional studies suggested that either there was a lack of interventional studies on multimorbidity or that interventional studies did not use any instruments to measure the level of multimorbidity."*

7) Why "the definition of high prevalence and significant impact would depend on the type of healthcare setting or geography?" If it is so, will the authors consider recommending different multimorbidity measure for different healthcare setting?

In another of our study looking at the patterns of multimorbidity, we have reported notable differences in the pattern of multimorbidity between studies from Asian and Western countries.

Due to the differing disease prevalence and impact in different healthcare settings or geographies, it would be good to develop a different multimorbidity measure for different healthcare settings. But this would be too confusing and too difficult for comparisons between different healthcare setting or geography. Therefore, we have amended in the Discussion section under 'Clinical Implications' segment (2nd-3rd paragraph, page 26-27) to make the message clearer.

"Ideally, a single instrument measuring the level of multimorbidity should be able to predict a variety of relevant outcomes. However, Byles et al. reported that a single instrument could not be used to predict different outcomes, in different patient groups and settings unless the instrument used different weights for the same items in calculating the scoring systems. Such multiple-scoring instruments may be the way forward for validation of prognostic models for different outcomes and different populations with established multimorbidity instruments. For example, depending on the outcome, study population and setting, the choice of conditions included in the multiple-scoring instrument should include those with a high prevalence in that study population and the weights should be determined by their significant impact (i.e. outcome) on the affected population.

For pragmatic reasons, the final selection of the conditions to be included in such a multiple-scoring instrument may still have to take into account the availability of relevant and reliable data. A certain degree of reductionism will also have to be accepted because a single instrument will not be able to encompass all the nuances of the different interactions of chronic conditions on an individual living in

his/her unique milieu. We recommend that researchers perform validation studies using the instruments listed in this systematic review to adjust the weights according to the specific outcome of interest for the study population relevant to their setting."

Minor comments:
8) Page 7 Line 17-19: We included instruments that measured multimorbidity for the purposes of establishing prevalence as well as instruments that measured multimorbidity for the purposes of predicting outcomes. Suggest changing "as well as instruments that measured multimorbidity for the purposes of" to "and".

This sentence was removed in the revision after the comment raised by the 2nd reviewer.

9) Page 7 Line 51, suggest changing "reported only instruments" to "only reported instruments".

Thank you for your suggestion. This was corrected and please refer to Methods section page 7.

10) Page 8 Line 24-26, suggest deleting (i.e., cross-sectional and cohort), since the two methods were mentioned earlier.

We have removed cross-sectional and cohort as suggested, and removed the mention of 'combined studies' as suggested by the 2nd reviewer.

Reviewer 2

Please leave your comments for the authors below:
1) This interesting systematic review examines contemporary instruments for measuring multimorbidity (the literature search was restricted to January 2010–August 2018), providing both a catalogue of these (from the abstract: "The main objective of this systematic review was to provide a list of instruments for measuring level of multimorbidity") along with assessments of each ("The secondary objective was to provide details of the requirements, strengths and limitations of these instruments"). Such a review would be useful to researchers, refreshing some of the information in Huntley, et al. (2012), particularly for readers who are not specialists in this area.

Thank you for your comment.

2) In order to help this group of researchers, I think that the systematic review needs to be refreshed. The searches were conducted over two years ago now and as you state in the registration "There will be the novel ones that have only been recently developed.", a

statement that is presumably as true now as it was in 2018. Similarly, in the background (Page 6, Line 15), you suggest exponential growth in the literature, suggesting that there could be a reasonable quantity of work published in those two years.

Thank you for your suggestion. We have refreshed and updated the search to 23rd Oct 2020.

3) It's not clear to me how the risk of bias is relevant here, and consequently why only studies with a low risk of bias were included in the review. It seems that 67 studies were identified and a third of these (21 studies) excluded due to a higher risk of bias; which in turn excluded one sixth, 6 out of a possible 33, of the instruments—non-trivial reductions in both studies and instruments. As Huntley, et al. (2012) noted, "we have not set out to assess individual study quality", which would have been even more relevant to them as they were interested in both reliability and validity. If you were looking at the results of the studies as your focus (which is not part of the objectives, as I read them), risk of bias assessment would be expected, but an instrument is not made less interesting because the researchers used inappropriate statistical tests or did not conceal allocation appropriately (which would increase the study's risk of bias and increase the chances that it was excluded on this basis here, irrespective of the quality of the instrument itself). Can you explain why the risk of bias assessment is needed here?

The aim of this review was to provide an overall recommendation of the list of instruments that have been appropriately used to determine an outcome. While the emphasis of this review was on the measure of multimorbidity, the risk of bias assessment gave equal weightage for the risk of selection, comparability and outcomes in the study design. Providing a list of instruments for measuring the level of multimorbidity without considering the quality of the study design would potentially wrongly associate an instrument with a clinically important outcome and mislead the reader. We have clarified that in the introduction section (5th paragraph, page 6). "*The main objective of this review was to list instruments for measuring the levels of multimorbidity where risk of bias of the study design was assessed to be low. We specifically look for studies that measure the association of a clinically important outcome with different levels of multimorbidity as the main independent variable in community-dwelling individuals."* and in the Methods section (4th paragraph, page 7-8) *"As the aim of this review was to provide a list of instruments for measuring the levels of multimorbidity where risk of bias of the study design was assessed, the selected articles were next appraised independently by three reviewers (ESL, EH and TSH) using the modified Newcastle-Ottawa Scale (NOS)[1 2]…"*

4) You also refer to the study needing at least one outcome (e.g. Page 3, Line 30; Page 6, Line 56; this is most clearly stated on Page 6, Lines 25–31: "We specifically looked at instruments that measure the prediction of a specific outcome with different levels of multimorbidity in community-dwelling individuals."), and this seems to refer to a dependent variable modelled using the multimorbidity instrument as an independent variable. While the ways that multimorbidity instruments are used and when they were statistically significantly associated with outcomes of interest (as shown on Page 8, Lines 37–43 and in Tables 1, 3a, and 3b) is

definitely interesting, although the strength/clinical relevance of the associations would need to be described if this was intended to assess instrument's validity, statistical significance alone would not suffice, if the present work is to serve as a catalogue guiding researchers in selecting an appropriate instrument to measure multimorbidity (as I interpret the study objectives), this seems an unnecessary exclusion. The approach taken here would also seem to rule our prevalence studies.

Thank you for your comment. Prevalence studies were actually excluded from the study. However, the sentence *"We included instruments that measured multimorbidity for the purposes of establishing prevalence as well as instruments that measured multimorbidity for the purposes of predicting outcomes"* was put in to emphasise that prevalence studies that also included an outcome variable were selected. In retrospect, we realise that this sentence may confuse the reader and we have decided to remove the sentence altogether.

Assessing the quality of an instrument/measure such as the psychometric properties was beyond the scope of this systematic review. We have put this in the Discussion section under 'Strengths and Limitations' segment (page 27). . "*This systematic review did not review the validity and reliability of all the instruments as it was beyond the scope of the intended work. We have, however, included the references of the original papers or validation studies in Table 2 where each of the instrument was described*."

We have also expanded the reporting of the outcomes and provide a brief discussion on their clinical relevance. Please see 'Outcomes' segment under Results section in page 22 and Discussion section in page 25. We also included "*No studies were excluded for an outcome that was not deemed to be clinically important*." at page 8 in the Results section.

5) Do you think that different measures of multimorbidity should be used depending on whether the outcome of interest is, for example, quality of life-based versus healthcare cost-based? That would suggest that there are multiple multimorbidity constructs that vary by outcome (not just by population of interest). I think that there is much more to what you say on Page 21, Lines 36–42 than readers might appreciate. Do you believe that the multimorbidity is a single construct, at least for a particular well-defined population, or are there multiple constructs even within a particular well-defined population?

Thank you for your comment and question on this section. Indeed, there is much more than what was originally stated. We have revised and improved on the section to make our message come across better. Please see Discussion section under Clinical Implications segment (2nd-3rd paragraph, page 26-7).

"Ideally, a single instrument measuring the level of multimorbidity should be able to predict a variety of relevant outcomes. However, Byles et al. reported that a single instrument could not be used to predict different outcomes, in different patient groups and settings unless the instrument used different weights for the same items in calculating the scoring systems.

Such multiple-scoring instruments may be the way forward for validation of prognostic models for different outcomes and different populations with established multimorbidity instruments. For example, depending on the outcome, study population and setting, the choice of conditions included in the multiple-scoring instrument should include those with a high prevalence in that study population and the weights should be determined by their significant impact (i.e. outcome) on the affected population.

For pragmatic reasons, the final selection of the conditions to be included in such a multiple-scoring instrument may still have to take into account the availability of relevant and reliable data. A certain degree of reductionism also have to be accepted because a single instrument will not be able to encompass all the nuances of the different interactions of chronic conditions on an individual living in his/her unique milieu. We recommend that researchers perform validation studies using the instruments listed in this systematic review to adjust the weights according to the specific outcome of interest for the study population relevant to their setting."

6) Similarly, why would it matter whether multimorbidity was "the main independent variable" and not "a covariate" (Page 7, Lines 5–9)? How would this be operationalised when a study investigates a range of potential variables where one of those was multimorbidity, alongside SES and ethnicity, say, and where all independent variables are of interest? Why would multimorbidity not be of interest if it was a mediator instead of the predictor of interest (say, looking at ethnicity differences in support for euthanasia and asking whether these are mediated by the extent or presence of multimorbidities)? This would seem to potentially penalise some more sophisticated or comprehensive studies, at least in theory, whereas simpler or piecewise publications of the same research could still be eligible. What about multimorbidity as an outcome?

This study specifically aimed to look at the association of the level of multimorbidity as the main independent variable, controlling for other covariates, with an outcome variable. We assessed every article to look in the introduction section whether the authors described and considered multimorbidity as one of the main independent variables. We believe that considering studies that include the level of multimorbidity as a mediating, confounding or effect-modifying variable will distract and include some studies that have not given much thought to the concept of multimorbidity and may include many studies that consider the concept of comorbidity instead without differentiating the subtle differences between the two.[3]

We have emphasised this to the reader in the main objective in Introduction section paragraph 5 page 6, and also in the Discussion section under 'strength and limitations of the study' segment paragraph 2 page 27.

"Finally, this review specifically aimed to look at the association of the level of multimorbidity as the main independent variable and excluded the level of multimorbidity as a mediating, confounding or effect-modifying variable. We believe that the strict criteria would reduce the number of studies included but will clearly exclude studies that does not differentiate the subtle differences between comorbidity and multimorbidity."

We were also very clear at the outset that we wanted to find out whether the outcomes reported for the selected instruments were in accordance to the set of core outcomes of multimorbidity (COSmm) as mentioned in paragraph two in the introduction. We have amended the second objective in the Background section to make this clearer (page 6) *"Our second objective was to provide details of the requirements, strengths and limitations of these instruments, and the chosen outcomes in the studies so that clinicians and researchers can select or develop instruments that match their needs."* Therefore, this systematic review did not consider looking at studies that considered multimorbidity as an outcome.

7) What I'm asking with all of these questions is perhaps more simply asked as whether you believe that multimorbidity should be measured in different ways for these different contexts? If so, it would seem more useful to me to stratify the reporting by these criteria rather than excluding studies and focusing on a rather specific use case. It seems very likely to me that readers interested in your work will wish to incorporate multimorbidity in a wide range of contexts (exposure, confounder, mediator, effect modifier, or outcome) and that by restricting your inclusion in this way, you are substantially reducing the possible use of your work. If, however, you are only interested in your work being of direct interest to the particular use case defined through your inclusion and exclusion criteria, that should be made very explicit (including perhaps in the title, certainly in the abstract) so that readers with other interests can continue their search elsewhere.

We sincerely appreciate your comments as it helps to clarify our thought processes which were not very well articulated. We have revised the title, made changes to the introduction in the abstract and also the first objective in the introduction.

Title (page 1): *A systematic review on the instruments used for measuring the association of the level of multimorbidity and clinically important outcomes*

Introduction section (5th paragraph, page 6): *The main objective of this review was to list instruments for measuring the levels of multimorbidity where risk of bias of the study design was assessed to be low. We specifically look for studies that measure the association of a clinically important outcome with different levels of multimorbidity as the main independent variable in community-dwelling individuals*.

Methods section (4th paragraph, page 7): *As the aim of this review was to provide a list of instruments for measuring the levels of multimorbidity where risk of bias of the study design was assessed, the selected articles were next appraised independently by three reviewers (ESL, EH and TSH) using the modified Newcastle-Ottawa Scale (NOS). Each article was assessed under the three broad categories: a) Selection; b) Comparability; and c) Outcome (Appendices 2 & 3). We only reported instruments that were described in studies with a low risk of bias.*

8) If I was a researcher considering instruments for assessing multimorbidity, for any purpose, reliability and validity (as Huntley, et al. looked at) would seem crucial. While you provide information relevant for some forms of validity, I didn't see any direct mention of the issue of reliability. Can you explain why this aspect of the instruments was not included in your review? Reliability coefficients would potentially be amenable to meta-analysis, which could be very helpful to readers. In any case, I think that these should fall under the heading of strengths and weaknesses of the instruments. The burden on participants, where appropriate (interviews) and reported, could be a useful addition to Table 2 for someone weighing up various instruments. The financial cost (see Page 20, Lines 13–17) of instruments would also seem to be of interest and could be easily added to Table 2.

Assessing the quality of an instrument/measure such as the psychometric properties was beyond the scope of this systematic review. We have put this in the Discussion section under 'Strengths and Limitations' segment, page 27. This systematic review did not review the validity and reliability of the list of instruments as it was beyond the scope of the intended work. We have, however, included the references of the original papers or validation studies in Table 2 where each of the instrument was described.

We have separately looked into all the 69 selected studies and confirmed that none of them reported the reliability of their scales.

Thank you for your suggestion to include the 'burden on participant' and financial cost of the instruments. We have made the necessary amendments to reflect that for the financial cost and put in 'participant involvement required' in Table 2.

9) In the registration you say that "We expect high heterogeneity in the selected studies and meta-analysis will not be attempted." but it's not clear to me how you would perform a meta-analysis here unless it was looking at reliability or at associations between multimorbidity and outcomes. However, the latter is not part of the study objectives based on my reading. What would such a meta-analysis have looked like otherwise, assuming that heterogeneity was not an issue?

Actually, we did not expect to be able to perform meta-analysis because of the expected variety of instruments and outcomes. Hypothetically, if there was low heterogeneity among the studies, we would have conducted a meta-analysis on the association on the most common multimorbidity instrument with the most common outcome.

10) I'm not clear as to why non-English language articles were excluded. Could you explain, including in the manuscript, why this approach was taken?

We excluded non-English-language articles due to lack of professional translators or trained research team members who can deal with non-English articles or have access to non-English specialised databases.

11) There are a few minor issues with language (e.g. "Studies published between January 2010 and August 2018 inclusive, in English language." in the abstract is not a complete sentence. Careful proofreading throughout should find and correct these and other small issues.

We have made the necessary changes and checked through the manuscript and corrected similar errors.

12) In the article summary, you say "This review builds upon Huntley et al.'s 2009 review paper". Do you mean their 2012 paper? Their search was up to December 2009, which might be your intended point here? See also Page 6, Lines 12–16 and Line 19, and perhaps elsewhere.

The correction has been done.

13) You mention (Page 5, Line 45) "a Delphi panel of experts" but Delphi is a method/process not a characteristic.

This has been changed to the following sentence:

(Page 5) *A set of core outcomes of multimorbidity (COSmm) was proposed after consulting a panel of international experts of multimorbidity intervention studies using a Delphi process.*

14) You say that you contacted authors where additional information or clarification was needed (Page 7, Line 57). How often did this occur? How often was it successful?

We have updated the information and included in the Methods section paragraph 5, page 7. "*We contacted the authors, as needed, for additional information or clarification up to three times spaced one week apart. We contacted 25 authors, and 19 of them replied."*

15) Page 8, Lines 21–27: I would think that a study that combined cross-sectional and cohort aspects would be likely to be a cohort study where analyses were also performed on the baseline measurements. If a study (as a study design) follows participants over time, it is a cohort study not a cross-section/cohort study; there may be cross-sectional and longitudinal analyses, but the study itself is a cohort study.

Thank you for pointing this out and we agree with your comments. We have made the necessary changes in Figure 1, and the Results section (1st paragraph, page 8).

16) I disagree that disease count "assumes a linear relationship on an ordinal scale" (Page 19, Lines 13–16). An ordinal scale is defined in terms of ordering not linearity, and any assumption of a linear relationship would not be a function of the instrument itself, but rather a researcher's use of the instrument through treating it as a continuous exposure variable. Depending on the taxonomy for measurement scales being used, counts are absolute in any case (not ordinal, interval, or ratio). There is nothing stopping a researcher from modelling non-linearities (e.g. through splines, higher-order polynomial terms, or fractional polynomials) or (aside from the reduction in statistical power) creating categories based on cut-points. None of these would "…[ignore] the fact that an additional unit of increase in 'disease count' will likely have a diminishing impact." (Lines 15–19). It is important to keep distinct the aspects of the instrument and aspects of how it could be used.

We thank you for your comments and teaching on this aspect of basic measurement knowledge. We have removed the sentence.

17) I don't think that it is surprising that only observational studies were found (although I'm not sure whether Page 21, Lines 13–18 is referring to pre- or post-exclusions). As multimorbidity was required to be the primary outcome, intervention studies would only be eligible if they manipulated multimorbidity, which would obviously not pass any ethical review. It seems to me that your inclusion and exclusion criteria themselves would prevent any intervention studies from being included. While Line 17–22 might well be true in general, your findings can't reflect this unless they are based on the lack of such studies identified pre-exclusions.

By intervention studies, we meant studies that intervene to improve on the management of multimorbidity in the clinical setting for example using a chronic care model, promotion of self-management, nurse-led case management etc. We were hoping that some of these studies would be selected that measured the outcomes for the different levels of multimorbidity before and after intervention, for example. These studies would be evaluation studies on the effectiveness of the intervention of patients with multimorbidity.

18) I didn't understand the logic of splitting Table 3 into 2 tables (one for positive associations and one for negative or no evidence of associations [not "no associations", see below]). As an example, wouldn't "deficits of perceived general health" (in Table 3a) be loosely the inverse of "general health", perceived health status", or "self-rated health" (in Table 3b column one)? Whether a relationship is positive or negative could also depend on how the dependent variable is coded (satisfaction 1=low and 5=high, versus the same with 1=high and 5=low, versus dissatisfaction with either NRS). Between these two points, I don't personally find this distinction between the two tables to be helpful. Other groupings, perhaps thematic, might be more useful though. I'd suggest caution in the "no association" part of Table 3b, as you are not claiming that there is no association, merely indicating that there was no evidence for such associations (and I'll point out that this is likely in some cases to depend on the other variables included in the model as well as the sample size).

We thank you for the feedback. We were initially worried that Table 3 would be too big and confusing. We have adopted your suggestion and combined them into one table. Yes, we agree that positive, negative and no associations were findings from the 69 studies only. We have amended the title of Table 3 to remind readers that the table reflected only findings from the 69 studies with low risk of bias.

In summary:

1) There doesn't seem to be any justification provided for requiring multimorbidity to be the primary exposure of interest. You might have a reason for this position, but it isn't clear to me. If this is part of the design of the study, it needs to be very clear to readers.

We have clarified this as mentioned in the responses to questions 6 and 7.

2) Similarly, I don't see the reason for the focus (and requirement) for outcome variables unless there is an argument and subsequent discussion about "horses for courses" in multimorbidity measurement.

We have clarified this as mentioned in the responses to question 4 and 6.

3) The risk of bias assessment didn't seem relevant to me.

We have clarified this as mentioned in the responses to question 3.

4) Some discussion around reliability, and more around validity and clinical significance of associations, could be extremely helpful to readers.

We have provided the references for the original studies in Table 2 for readers to refer to the reliability and validity of the instrument. We have also added more information in Table 1 and a section of results for prediction models for the validity. Finally, we have discussed the clinical significance of the associations under Outcomes in Discussion section at page 25.

5) I think that the search needs to be refreshed to the present date.

This has been updated to 23rd Oct 2020.

VERSION 2 – REVIEW

REVIEWER	Andrew Robert Gray Biostatistics Centre, Division of Health Sciences, University of Otago, New Zealand
REVIEW RETURNED	21-Jan-2021

GENERAL COMMENTS	Thank you for your responses to my queries, and for the revisions that these prompted. I'm delighted to see the search updated, with, I think, an additional 29 studies and six new instruments and I appreciate the additional details and improved clarity throughout the manuscript. However, I'm sorry, but I still don't understand the reason for only including studies with a low risk of bias. I see that the other reviewer

	also raised this point and I carefully read your response to their question as well as your response to mine. You say that “The aim of this review was to provide an overall recommendation of the list of instruments that have been appropriately used to determine an outcome.” in your response and I understand the text you have added: “The main objective of this review was to list instruments for measuring the levels of multimorbidity where risk of bias of the study design was assessed to be low.” However, reading the new manuscript and your responses as a whole, I still don’t understand why you would not recommend an instrument simply because it was only used in a study, or set of studies, with non-low risk of bias. You explain that “[p]roviding a list of instruments for measuring the level of multimorbidity without considering the quality of the study design would potentially wrongly associate an instrument with a clinically important outcome and mislead the reader.” but this doesn’t seem to directly address my comments, e.g. that an article using an inappropriate statistical method in their regression analyses wouldn’t seem to rule out the instrument that they used. It would help if you could give some specific examples of where you feel the inclusion of these other studies might “mislead the reader”. As the first reviewer noted, and I agree, a high quality study could use a low quality instrument, and a low quality study could use a high quality instrument. Your response “By intervention studies, we meant studies that intervene to improve on the management of multimorbidity in the clinical setting for example using a chronic care model, promotion of self-management, nurse-led case management etc. We were hoping that some of these studies would be selected that measured the outcomes for the different levels of multimorbidity before and after intervention, for example. These studies would be evaluation studies on the effectiveness of the intervention of patients with multimorbidity.” would seem to require that multimorbidity was either an outcome variable (and so would be excluded) or a not-main-independent variable (and so excluded). If there was a pre-post design, time would be the main independent variable with multimorbidity either another independent variable or an effect modifier of time; if there were parallel groups, the group assignment would be the main independent variable. If analyses were conducted before and after the intervention only, where multimorbidity could be the main independent variable, these would be two separate cross-sectional analyses rather than an intervention study. My apologies if I’m missing some subtle (to me) point here, but by requiring multimorbidity to be the main independent variable, it seems to me that you are, by definition, excluding all intervention studies entirely, making your observation on Page 26, Lines 12–20 inevitable rather than something that could be concluded based on the data you have collected. Can you explain more precisely how you think an intervention study could qualify under your current inclusion/exclusion criteria and given an example (real or otherwise) to illustrate this? I appreciate that you clearly explain that you have excluded studies where an outcome variable was not examined for its association with multimorbidity (a pure prevalence study, as I would use the term and which was apparently eligible, wouldn’t look for such associations, so perhaps we are using the term “prevalence study” in different ways here?), but I still don’t understand the why question there either. Can you give some specific scenarios where you feel that a reader could be misled if these study designs were included? You

	have made it very clear what you have done, and I thank you for that, but I still don't understand why you have done this. I do agree with you that some measures claiming to be of multimorbidity might be instead measures of comorbidity, but that would seem to be a quality of the instrument (its face and content validity at least) rather than of the study. Given what is reported here, surely it is only the quality of the instrument that is (ultimately) important, and given that no assessment of the instruments' quality was performed, I don't see how the study risk of bias is important—unless you are assuming that low quality studies will always use low quality instruments and that high quality studies will always use high quality instruments. If this is the motivation, this is a very long bow to draw. You have made what you are doing very clear in your manuscript, so my question here is about why you are doing this. You could add some theoretical or empirical argument for your chosen approach, but I think the simplest approaches are still to include all instruments from all studies and perhaps to indicate the study quality and the manner in which the multimorbidity measure was used alongside each, as you do for study design, etc. I think that it's entirely plausible that a measure of multimorbidity being used as a confounder could have received less thought and consideration than where it is the main independent variable but this would be only on average. If that is the argument being used here, should all studies published in low impact journals (say Q3 and Q4 in JCR) be removed as the same would, on average, apply there; and should all studies that have not been cited at an appropriate rate be excluded for the same reason? I would expect that a measure of multimorbidity used as a confounder in a study published in the BMJ or the NEJM would have been considered and scrutinized more, not less, on average than a measure of multimorbidity used as the main independent variable in a low impact journal. At the risk of further complicating this point, there is also the risk that study questions evolved over time. We all know that some studies set out to look at pre-identified dependent and independent variables to answer a specific research question as their focus and end up reporting on other questions, and so different variables, sometimes based on statistical significance. This situation can be due to departures from the initial analysis plan or is a feature of secondary analyses of collected data. However, how could we know whether the role of multimorbidity didn't shift from the original planning of the study through to the publication? This seems particularly important in secondary analyses where a poorly conceived measure of multimorbidity could have been constructed, for example as a potential confounder, and then investigated as the main independent variable in a later publication. It would be a huge undertaking to distinguish between what was planned and what was reported to have been done and practically impossible for unregistered studies. Overall, I don't believe that it's practical, if even possible, to distinguish clearly between well-conceived and well-scrutinized instruments based on the resulting publication and I can't see any other reason for excluding some studies, although I am open to being persuaded on this point. Based on Figure 1, it looks like around 20 articles would be added back in if how multimorbidity was used was ignored, and 27 if RoB was ignored. It seems plausible that new instruments would emerge from these on top of those from

	the 69 you've already examined. In the absence of such arguments, though, I'm afraid that my suggestion remains that all studies where multimorbidity measures were used should be included, irrespective of the study design, study quality, and role of the multimorbidity measure in the descriptive and/or inferential statistical analyses. Describing which instruments are used for different designs, in higher or lower quality articles, and by purpose would be interesting but I don't think exclusions can be reliably made on these bases. I'd be very happy to consider specific examples and/or hypothetical scenarios where including studies that are low quality or where the multimorbidity measure was used other than as the "main independent variable" (including when it is one of several exposures of interest) would mislead or distract the reader as I honestly can't think of any myself. If you believe that multimorbidity should be measured differently depending on the criteria you have used (study quality, how multimorbidity is measured, and [by implications] study design), which you have revised the title, abstract, and objectives to indicate, I think you need to clearly justify these exclusions, as well as describing them, as you do. Again, and I apologise for the repetition, I appreciate what you have done, but I'm still not able to understand why. As a separate comment, I appreciate the clarification about not performing a meta-analysis on Page 8, Lines 5–9, but this seems to leave the question of what would you have subjected to meta-analysis had there been an expectation of homogeneity (and in what your a prior expectation of heterogeneity was). I don't see any issues with a systematic review that was never intended to look at a meta-analysis, the literature contains many of these, so perhaps this sentence should be deleted unless sufficient clarification can be added. Finally, I see that Table 3 has been combined, but still uses the headings of "positive association", "negative association" and "no association". My comments about these divisions from before still stand. The direction of the association here would still seem to depend on how the outcome is coded (some of the variables under "positive association" seem to be the inverse of a variable under "negative association", at least such that a solid negative correlation would be expected between them a priori to examining any data. "no association" still seems to me an absolute claim rather than describing a lack of evidence for an association. It might be worth rereading my comment number 18 from my previous review in light of this. I think this manuscript could be a valuable resource to someone looking at measuring multimorbidity in their study, irrespective of how they were intending to use this construct, and you've done well in extending the data collection period. The challenge, as I see it, is to either justify the exclusions you've made or add these additional studies, perhaps further describing the instruments by quality and the purpose of the multimorbidity variable.

VERSION 2 – AUTHOR RESPONSE

1. Thank you for your responses to my queries, and for the revisions that these prompted. I'm delighted to see the search updated, with, I think, an additional 29 studies and six new instruments and I appreciate the additional details and improved clarity throughout the manuscript.

    - *Thank you for your time in reviewing our manuscript and providing invaluable comments for improvement.*

2. However, I'm sorry, but I still don't understand the reason for only including studies with a low risk of bias. I see that the other reviewer also raised this point and I carefully read your response to their question as well as your response to mine. You say that "The aim of this review was to provide an overall recommendation of the list of instruments that have been appropriately used to determine an outcome." in your response and I understand the text you have added: "The main objective of this review was to list instruments for measuring the levels of multimorbidity where risk of bias of the study design was assessed to be low." However, reading the new manuscript and your responses as a whole, I still don't understand why you would not recommend an instrument simply because it was only used in a study, or set of studies, with non-low risk of bias. You explain that "[p]roviding a list of instruments for measuring the level of multimorbidity without considering the quality of the study design would potentially wrongly associate an instrument with a clinically important outcome and mislead the reader." but this doesn't seem to directly address my comments, e.g. that an article using an inappropriate statistical method in their regression analyses wouldn't seem to rule out the instrument that they used. It would help if you could give some specific examples of where you feel the inclusion of these other studies might "mislead the reader". As the first reviewer noted, and I agree, a high quality study could use a low quality instrument, and a low quality study could use a high quality instrument.

    - *We have reflected on the above comments and agree that there is no good reason for listing only instruments that are found in studies with a low risk of bias. We have added the instruments from studies rated as high risk of bias into Table 1, 2 and 3. With the inclusion of all the studies (irrespective of the risk of bias appraisal), we identified an additional instrument, which is the Disease Burden Impact Scale (DBIS).*

3. Your response "By intervention studies, we meant studies that intervene to improve on the management of multimorbidity in the clinical setting for example using a chronic care model, promotion of self-management, nurse-led case management etc. We were hoping that some of these studies would be selected that measured the outcomes for the different levels of multimorbidity before and after intervention, for example. These studies would be evaluation studies on the effectiveness of the intervention of patients with multimorbidity." would seem to require that multimorbidity was either an outcome variable (and so would be excluded) or a not-main-independent variable (and so excluded). If there was a pre-post design, time would be the main independent variable with multimorbidity either another independent variable or an effect modifier of time; if there were parallel groups, the group assignment would be the main independent variable. If analyses were conducted before and after the intervention only, where multimorbidity could be the main independent variable, these would be two separate cross-sectional analyses rather than an intervention study. My apologies if I'm missing some subtle (to me) point here, but by requiring multimorbidity to be the main independent variable, it seems to me that you are, by definition, excluding all intervention studies entirely, making your observation on Page 26, Lines 12–20 inevitable rather than something that could be concluded based on the data you have collected. Can you explain more precisely how you think an intervention study could qualify under your current inclusion/exclusion criteria and given an example (real or otherwise) to illustrate this?

    - *Thank you for pointing out very clearly how we have inadvertently excluded intervention studies entirely using the current inclusion and exclusion criteria. However, we have also not come across any intervention studies during our search and have, thus, included that sentence initially.*

- *We have deleted the paragraph "Although there was no restriction on the type of study design in our search strategy, we only found observational studies in our search. The lack of interventional studies suggested that either there was a lack of interventional studies on multimorbidity or that interventional studies did not use any instruments to measure the level of multimorbidity." in page 26, lines 12-20 as it may confuse the reader.*

4. I appreciate that you clearly explain that you have excluded studies where an outcome variable was not examined for its association with multimorbidity (a pure prevalence study, as I would use the term and which was apparently eligible, wouldn't look for such associations, so perhaps we are using the term "prevalence study" in different ways here?), but I still don't understand the why question there either. Can you give some specific scenarios where you feel that a reader could be misled if these study designs were included? You have made it very clear what you have done, and I thank you for that, but I still don't understand why you have done this. I do agree with you that some measures claiming to be of multimorbidity might be instead measures of comorbidity, but that would seem to be a quality of the instrument (its face and content validity at least) rather than of the study.

    - *According to de Vet et al., the three main purposes of measurement in medicine are for diagnosis, evaluation of intervention and prediction of outcome. (de Vet HCW, Terwee CB, Mokkink LB, et al. Chapter 3 - Development of a measurement instrument. Measurement in Medicine: A Practical Guide. Cambridge: Cambridge University Press 2011.) Discriminant measurement is for diagnosis, evaluation measurement is for evaluation after an intervention, and predictive measurement is to predict a specific outcome. We were hoping that the systematic review could help identify clinically meaningful outcomes for clinicians where the purpose of measuring the level of multimorbidity was based upon. The ultimate aim for providing a list of instruments for measuring the level of multimorbidity was for readers to ponder whether the instrument(s) was useful for them to predict the outcome of interest in their clinical practice. We are not so much interested in discriminant measurement whereby the levels of multimorbidity were described in a population with no specific outcome in mind. I hope this helps to explain the rationale of choosing an outcome.*

    - *We have amended the conclusion in the abstract to make this point clearer. "We listed the details of 33 instruments for measuring the level of multimorbidity as a resource for investigators interested in the measurement of multimorbidity for its association or prediction of a specific outcome."*

5. Given what is reported here, surely it is only the quality of the instrument that is (ultimately) important, and given that no assessment of the instruments' quality was performed, I don't see how the study risk of bias is important—unless you are assuming that low quality studies will always use low quality instruments and that high quality studies will always use high quality instruments. If this is the motivation, this is a very long bow to draw. You have made what you are doing very clear in your manuscript, so my question here is about why you are doing this. You could add some theoretical or empirical argument for your chosen approach, but I think the simplest approaches are still to include all instruments from all studies and perhaps to indicate the study quality and the manner in which the multimorbidity measure was used alongside each, as you do for study design, etc.

    - *Thank you for your comments, we have added all instruments into our manuscript and also indicated the assessment of the risk of bias in the last column of Table 1.*

6. I think that it's entirely plausible that a measure of multimorbidity being used as a confounder could have received less thought and consideration than where it is the main independent variable but this would be only on average. If that is the argument being used here, should all studies published in

low impact journals (say Q3 and Q4 in JCR) be removed as the same would, on average, apply there; and should all studies that have not been cited at an appropriate rate be excluded for the same reason? I would expect that a measure of multimorbidity used as a confounder in a study published in the BMJ or the NEJM would have been considered and scrutinized more, not less, on average than a measure of multimorbidity used as the main independent variable in a low impact journal.

At the risk of further complicating this point, there is also the risk that study questions evolved over time. We all know that some studies set out to look at pre-identified dependent and independent variables to answer a specific research question as their focus and end up reporting on other questions, and so different variables, sometimes based on statistical significance. This situation can be due to departures from the initial analysis plan or is a feature of secondary analyses of collected data. However, how could we know whether the role of multimorbidity didn't shift from the original planning of the study through to the publication? This seems particularly important in secondary analyses where a poorly conceived measure of multimorbidity could have been constructed, for example as a potential confounder, and then investigated as the main independent variable in a later publication. It would be a huge undertaking to distinguish between what was planned and what was reported to have been done and practically impossible for unregistered studies.

Overall, I don't believe that it's practical, if even possible, to distinguish clearly between well-conceived and well-scrutinized instruments based on the resulting publication and I can't see any other reason for excluding some studies, although I am open to being persuaded on this point. Based on Figure 1, it looks like around 20 articles would be added back in if how multimorbidity was used was ignored, and 27 if RoB was ignored. It seems plausible that new instruments would emerge from these on top of those from the 69 you've already examined.

In the absence of such arguments, though, I'm afraid that my suggestion remains that all studies where multimorbidity measures were used should be included, irrespective of the study design, study quality, and role of the multimorbidity measure in the descriptive and/or inferential statistical analyses. Describing which instruments are used for different designs, in higher or lower quality articles, and by purpose would be interesting but I don't think exclusions can be reliably made on these bases. I'd be very happy to consider specific examples and/or hypothetical scenarios where including studies that are low quality or where the multimorbidity measure was used other than as the "main independent variable" (including when it is one of several exposures of interest) would mislead or distract the reader as I honestly can't think of any myself.

If you believe that multimorbidity should be measured differently depending on the criteria you have used (study quality, how multimorbidity is measured, and [by implications] study design), which you have revised the title, abstract, and objectives to indicate, I think you need to clearly justify these exclusions, as well as describing them, as you do. Again, and I apologise for the repetition, I appreciate what you have done, but I'm still not able to understand why.

- *Thank you for explaining this very subtle but important concept about the difference between the main independent variable and that of the covariates used. We appreciate the effort you have put in to explain this very important concept clearly. Unfortunately, the study has started out without this thought in mind.*

- *As such, we have gone back to the 17 rejected articles due to "measurement of multimorbidity is a covariate only" and listed down all the instruments used for measuring multimorbidity. There was no new instrument identified that has not already been listed in the systematic review.*

- *We have also included this as a limitation of the study as follows:*

- *Finally, this review specifically aimed to look at the association of the level of multimorbidity as the main independent variable and excluded the level of multimorbidity as a mediating, confounding or effect-modifying variable. This strict criterion excluded 17 studies (Figure 1) as a result. Excluding these 17 studies did not alter the findings as the instruments used in all the 17 studies were Disease Count (n=9), Cumulative Illness Rating Scale (CIRS, n=3), Charlson Comorbdity Index (CCI, n=3), Elixhauser Index (EI, n=1) and Aggregated Diagnosis Groups (ADG, n=1) where no new instruments were identified.*

7. As a separate comment, I appreciate the clarification about not performing a meta-analysis on Page 8, Lines 5–9, but this seems to leave the question of what would you have subjected to meta-analysis had there been an expectation of homogeneity (and in what your a prior expectation of heterogeneity was). I don't see any issues with a systematic review that was never intended to look at a meta-analysis, the literature contains many of these, so perhaps this sentence should be deleted unless sufficient clarification can be added.

    - *Thank you for your comment and we have deleted the sentence.*

8. Finally, I see that Table 3 has been combined, but still uses the headings of "positive association", "negative association" and "no association". My comments about these divisions from before still stand. The direction of the association here would still seem to depend on how the outcome is coded (some of the variables under "positive association" seem to be the inverse of a variable under "negative association", at least such that a solid negative correlation would be expected between them a priori to examining any data. "no association" still seems to me an absolute claim rather than describing a lack of evidence for an association. It might be worth rereading my comment number 18 from my previous review in light of this.

    - *We apologise for missing this out. We have amended the headings of the table to "Evidence of an association", where any association to the outcome is grouped under this column and "No evidence of an association".*

I think this manuscript could be a valuable resource to someone looking at measuring multimorbidity in their study, irrespective of how they were intending to use this construct, and you've done well in extending the data collection period. The challenge, as I see it, is to either justify the exclusions you've made or add these additional studies, perhaps further describing the instruments by quality and the purpose of the multimorbidity variable.

- *Thank you for helping us to improve this systematic review. We have also tidied a few things up like using the term 'studies' or 'articles' only for describing the 96 studies instead of using 'papers' or 'records' at times. We have also cleaned up the tables to make it consistent for Table 2 and 3, and corrected some references that were cited wrongly.*

VERSION 3 – REVIEW

REVIEWER	Andrew Robert Gray Biostatistics Centre, Division of Health Sciences, University of Otago, New Zealand
REVIEW RETURNED	05-Apr-2021

GENERAL COMMENTS	Thank you for your responses and revisions. Well done on addressing my comments, which I must admit were challenging. I appreciate that this must have involved considerable work on your part and I think that you have strengthened this manuscript so that I would personally find it very useful, and at some stage presumably will find it useful, when considering measures of multimorbidity. I

	should also apologise for my very slow review. I hope that this delay has not been too inconvenient for you and that you are all well during these challenging times. There are perhaps a few places where the wording could be refined a little and I'll make some suggestions of these below, but you'll be pleased to know that I have no substantial comments to make. Abstract: Perhaps "The main objective of this systematic review was to provide a list of instruments that ARE SUITABLE FOR USE IN STUDIES AIMING TO measure the association of a specific outcome with different levels of multimorbidity as the main independent variable in community-dwelling individuals." as the instruments don't measure the association per se. Abstract: "published in English language" in the methods should presumably be connected to the database searches rather than the hand search of the Journal of Comorbidity. Abstract: Perhaps "We listed the details of 33 instruments for measuring the level of multimorbidity as a resource for investigators interested in the measurement of multimorbidity for its association WITH or prediction of a specific outcome." in the conclusion section. In the Discussion you say "last publication using DUSOI was in 2004" and I'm always wary of making absolute claims which could, in theory, be disproven with a single counterexample. I suggest qualifying this as something like: "TO OUR KNOWLEDGE, THE last publication using DUSOI was in 2004" so that a reader finding a publication, perhaps not in English or in a particularly obscure journal, after this date has not disproven your claim.

VERSION 3 – AUTHOR RESPONSE

Reviewer: 2
Mr. Andrew Gray, University of Otago
Comments to the Author:
Thank you for your responses and revisions. Well done on addressing my comments, which I must admit were challenging. I appreciate that this must have involved considerable work on your part and I think that you have strengthened this manuscript so that I would personally find it very useful, and at some stage presumably will find it useful, when considering measures of multimorbidity. I should also apologise for my very slow review. I hope that this delay has not been too inconvenient for you and that you are all well during these challenging times.

Thank you for your time, effort and patience in guiding us to improve the manuscript. We are very appreciative of your rigour and professionalism. We are keeping well in both Singapore and Canada. We hope you are too.

There are perhaps a few places where the wording could be refined a little and I'll make some suggestions of these below, but you'll be pleased to know that I have no substantial comments to make.

Abstract: Perhaps "The main objective of this systematic review was to provide a list of instruments that ARE SUITABLE FOR USE IN STUDIES AIMING TO measure the association of a specific

outcome with different levels of multimorbidity as the main independent variable in community-dwelling individuals." as the instruments don't measure the association per se.

Abstract: "published in English language" in the methods should presumably be connected to the database searches rather than the hand search of the Journal of Comorbidity.

Abstract: Perhaps "We listed the details of 33 instruments for measuring the level of multimorbidity as a resource for investigators interested in the measurement of multimorbidity for its association WITH or prediction of a specific outcome." in the conclusion section.

In the Discussion you say "last publication using DUSOI was in 2004" and I'm always wary of making absolute claims which could, in theory, be disproven with a single counterexample. I suggest qualifying this as something like: "TO OUR KNOWLEDGE, THE last publication using DUSOI was in 2004" so that a reader finding a publication, perhaps not in English or in a particularly obscure journal, after this date has not disproven your claim.

Thank you for the suggestions. They are all relevant and improve the clarity of the writing. We have made all the necessary changes and made some minor edits in the abstract to keep within the 300 word-limit.

Reviewer: 2
Competing interests of Reviewer: None.